AI101 Mini-Project · Team 80 · IIT Ropar

Diamond Price Prediction

Linear Regression on carat, cut, colour and clarity

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Repo: github.com/Vansh-Ahujaa/AI101-team-80-project

Problem and data

What and why

Predict a diamond's price from four simple traits: carat, cut, colour and clarity. A fair estimate helps buyers and sellers compare stones quickly.

Dataset source

https://www.kaggle.com/datasets/ayeshaseherr/diamonds

Columns in the data

carat | cut | color | clarity

price

depth | table | x | y | z

blue = input · amber = target · grey = not used

53,940
rows (diamonds)

11
columns

4 → 1
features → price

Approach

1 Load

diamonds.csv with pandas, 53,940 rows

2 Preprocess

One-hot encode cut, colour, clarity

3 Split

80/20: 43,152 train, 10,788 test

4 Train

Linear Regression (scikit-learn)

5 Evaluate

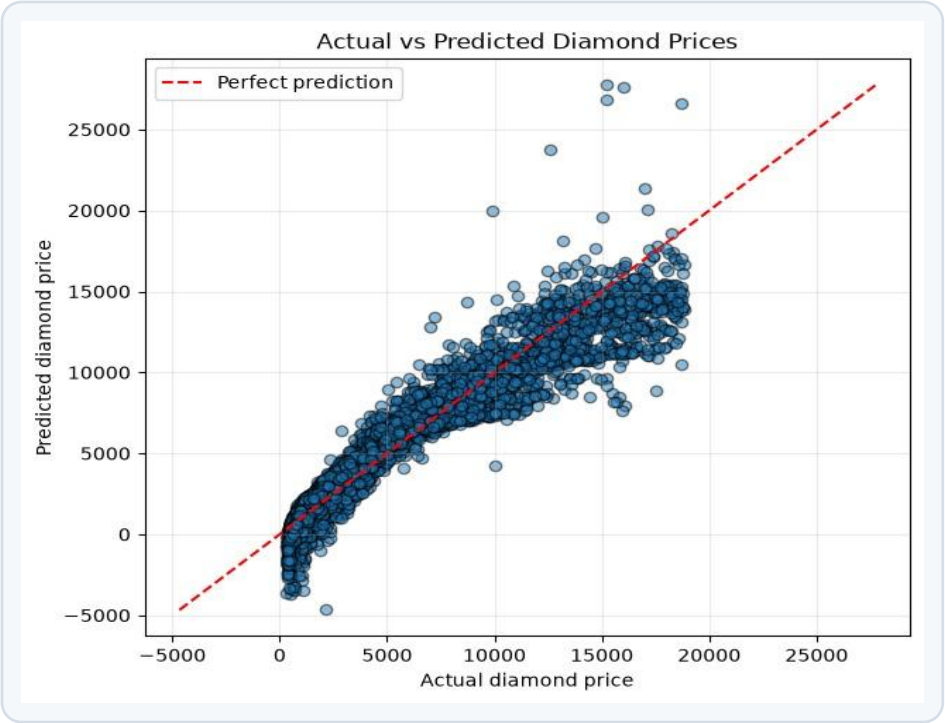
MAE, MSE, RMSE, R^2 and a plot

Libraries used

pandas **numpy** **matplotlib** **scikit-learn**

Model: price = b0 + b1·x1 + … + bn·xn, where x are carat and the encoded cut, colour and clarity columns

Results



Key numbers

R^2	MAE	RMSE
0.915	802	1,160

Demo: predicting a new diamond

0.75 carat · Good · F · VS2

$4,514

predicted price

Learnings

What worked

- No missing values in the data
- One-hot encoding made text columns usable
- Clear 80/20 split to test on unseen diamonds

What didn’t

- Prices are right-skewed: few costly stones
- Spread grows with carat, so a line fits less well
- 20 rows had a zero x, y or z

Who did what

- Vansh: run and setup commands, data analysis
- Bhuwan: created the readme with expected outputs, preprocessing of data
- Kavya: Made and edited the video, trained and tested the model
- Manthan: Made ppt, evaluated the model
- All four wrote and committed code